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Chapter 1: Number and Letter Series

Topic-wise short notes with solved examples

How to find the rule behind a number or letter series, with the common patterns, worked examples and practice.

1 Reading a number series

- A **series** is a list of numbers that follows one rule. Find the rule from the first few terms, then use it to find the missing term.
- First write the **differences** between neighbouring terms. If they are equal, the series is arithmetic.
- If the differences are not equal, try the **ratios**, then the differences of the differences.
- Always check the rule on **every** term you were given, not only the first two.

Example 1.1 Find the next term: 4, 9, 14, 19, ?

SOLUTION

The differences are $5, 5, 5$.

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2 Common patterns

Pattern	Example	Rule
Constant difference	7, 11, 15, 19	add 4 each time
Constant ratio	3, 6, 12, 24	multiply by 2
Squares	1, 4, 9, 16, 25	n^2
Cubes	1, 8, 27, 64	n^3
Square plus or minus one	2, 5, 10, 17, 26	$n^2 + 1$
Growing differences	1, 2, 4, 7, 11	add 1, then 2, then 3, then 4
Sum of the last two	1, 1, 2, 3, 5, 8	each term is the sum of the two before it
Primes	2, 3, 5, 7, 11, 13	the prime numbers in order

Example 2.1 Find the next term: 2, 5, 10, 17, 26, ?

SOLUTION

The terms are $1^2+1,\ 2^2+1,\ 3^2+1,\ 4^2+1,\ 5^2+1$.

The next is 6^2+1.

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Example 2.2 Find the next term: 3, 4, 6, 9, 13, ?

SOLUTION

The differences are $1, 2, 3, 4$, so the next difference is 5.

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3 Two series mixed together

- If the terms look unrelated, read the **odd places** and the **even places** as two separate series.
- Another trick: two operations used one after the other, such as "multiply by 2, then add 1".

Example 3.1 Find the next term: 2, 5, 4, 10, 8, 15, ?

SOLUTION

The terms in odd places are $2, 4, 8$ and are doubled each time, so the next is 16.

The terms in even places are $5, 10, 15$.

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Example 3.2 Find the missing term: 1, 3, 7, 15, 31, ?

SOLUTION

Each term is double the one before it, plus 1.

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4 Letter series

- Give every letter its **position**: A = 1, B = 2, ..., Z = 26. Then treat the series as a number series.
- The letter at position n from the front is the letter at position $27-n$ from the back: A and Z, B and Y, C and X, and so on.
- Letter groups usually follow one rule for the first letters, and another rule for the next letters.

Example 4.1 Find the next term: B, E, H, K, ?

SOLUTION

The positions are $2, 5, 8, 11$, so each step is $+3$. The next position is 14.

N

Example 4.2 Find the next term: AZ, BY, CX, ?

SOLUTION

The first letters move forward A, B, C, D. The second letters move backward Z, Y, X, W.

DW

REMEMBER

Write the position numbers under the letters before you look for the rule. It turns every letter series into a number series.

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